5. A Pleiotropic gene
(AIPMT 2015 – Re-exam)
a) Controls multiple traits in an individual
b) Is expressed only in primitive plants
c) Is a gene evolved during Pliocene
d) Controls a trait only in combination with another L gene

6. A true breeding plant is
(NEET Phase II 2016)
a) Near homozygous and produces offspring of its own kind
b) Always homozygous recessive in its genetic construction
c) One that is able to breed on its own
d) Produced due to cross pollination among unrelated plants

7. Mendel obtained wrinkled seeds in pea due to the deposition of sugars instead of starch. It was due to which enzyme?
(AIPMT 2001)
a) Amylase b) Invertase c) Diastase
d) **Absence of starch branching enzyme**

8. Ratio of complementary gene is
(AIPMT 2001)
a) 9:3:4 b) 12:3:1 c) 9:3:3:4 d) **9:7**

9. If there are 999 bases in an RNA that codes for a protein with 333 amino acid and the base at position 901 is deleted such that the length of the RNA becomes 998 bases, how many codons will be altered?
(NEET 2017)
a) 1 b) 11 c) 33 d) **333**

10. If a homozygous red flowered plant is crossed with a homozygous white flowered plant, then the off-springs will be
(AIIMS 1999, 2002, 2007)
a) Half-white flowered b) Half-red flowered
c) All white flowered d) **All red flowered**

11. The ratio in a dihyrbid test cross between two individuals is given by (AIIMS 2001)
a) 2:1 b) 1:2:1 c) 3:1 d) **1:1:1:1**

12. Pure line breed refers to
(AIIMS 2002, AIIMS 2007)
a) Heterozygosity only
b) Heterozygosity and linkage
c) **Homozygosity only**
d) Homozygosity and self assortment

13. How many different types of gametes can be formed by F_1 progeny, resulting from the following cross AABBCC x aabbcc
(AIIMS 2004)
a) 3 b) **8** c) 27 d) 64

14. Which of the following conditions represents a case of co-dominant genes?
(AIIMS 2009)
a) A gene expresses itself, suppressing the phenotypic effect of its alleles
b) Genes that are similar in phenotypic effect when present separately, but when together interact to produce a different trait
c) Alleles both of which interact to produce a trait which may or may not resemble either of the parental type
d) Alleles, each of which produces an independent effect in a heterozygous condition.

15. If 'A' represents the dominant gene and 'a' represents its recessive allele, which of the following would be most likely result in the first generation off spring when Aa is crossed with aa? (AIIMS 2016)
a) All will exhibit dominant phenotype
b) All will exhibit recessive phenotype
c) **Dominant and recessive phenotypes will be 50% each**
d) Dominant phenotype will be 75%

16. In *Pisum Sativum*, there are 14 chromosomes. How many types of homologous pairs can be prepared? (JIPMER 2010)
a) 14 **b) 7** c) 2^{14} d) 2^{10}

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